

Scientific Computation

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October 27, 2025

Lecture 18 - Computing Eigenvalues

Lecture Outline

- The Power method
- The Rayleigh-Ritz quotient
- The Spectral theorem
- The Bauer-Fike bound

A Variational Characterization of Eigenpairs

We can consider an optimization or *variational* view of eigenvalues. Consider the following optimization problem:

$$\min_{x \neq 0} \frac{x^T A x}{\|x\|^2} \Leftrightarrow \min_{\|x\|_2 = 1} x^T A x$$

Lagrangian: $\mathcal{L}(x, \lambda) = x^T A x + \lambda (\|x\|_2^2 - 1)$

$$\nabla_x \mathcal{L}(x, \lambda) = 2Ax + \lambda \frac{x}{\|x\|_2} = 0 \Rightarrow Ax = -\frac{\lambda}{2\|x\|_2} x$$

$\Rightarrow x^*$ (minimizer) is an eigenvector of A , so we choose the algebraically smallest eigenvalue and its eigenvector (λ, x_1) :

$$x_1^T A x_1 = x_1^T \lambda x_1 = \lambda_1 \|x_1\|^2 = \lambda_1 \leq \lambda \quad \forall \text{ other eigenvalues } \lambda \in \sigma(A)$$

The Rayleigh-Ritz Quotient

For any vector $x \neq 0$, we define the objective function of that optimization problem as the Rayleigh-Ritz quotient:

$$R_A(x) \triangleq \frac{x^T A x}{\|x\|^2}.$$

If x is any eigenvector with eigenvalue λ , then $R_A(x)$

$$= \frac{x^T A x}{\|x\|^2} = \frac{x^T \lambda x}{\|x\|^2} = \frac{\lambda \|x\|^2}{\|x\|^2} = \lambda$$

If v is an approximation to an eigenvector x with eigenvalue λ , then $R_A(v) \approx \lambda$

Power Method Uses the Rayleigh-Ritz Quotient

In summary, the power method for finding the largest (in magnitude, i.e. $|\cdot|$) eigenvalue is:

Input: initial guess v_0 for an eigenvector, matrix A

for $k=1, 2, \dots, k_{\max}$

$$\tilde{v}_k = A v_{k-1}$$

$$v_k = \frac{\tilde{v}_k}{\|\tilde{v}_k\|_2}$$

$$\lambda_1^{(k)} = R_A(v_k) = \frac{v_k^T A v_k}{\|v_k\|_2^2} = v_k^T A v_k$$

Termination criterion? $\hookrightarrow$

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Spectral Theorem for Real Symmetric Matrices

Recall that if $A \in \mathbb{R}^{n \times n}$ is symmetric, we have real eigenvalues. Furthermore, the Spectral Theorem tells us that in the diagonalization

$$A = Q \Lambda Q^{-1}, \quad \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

Then $Q^{-1} = Q^T$, i.e. Q is an orthogonal matrix ($Q^T Q = I$). This is good because transposing a matrix is essentially "free" compared to inverting Q contains eigenvectors as columns.

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Key Property in Proving the Spectral Theorem

We have already seen that real symmetric matrices have real eigenvalues. How is the orthogonality of the eigenvectors proved?

Consider 2 distinct eigenvalues $\lambda_1 \neq \lambda_2$, and their eigenvectors x_1 and x_2 .

$$x_1^T A x_2 = x_1^T \lambda_2 x_2 = \lambda_2 (x_1^T x_2)$$

$$\xrightarrow{\text{Scalar}} x_2^T A^T x_1 = x_2^T A x_1 = x_2^T \lambda_1 x_1 = \lambda_1 (x_2^T x_1)$$

$$\lambda_2 (x_1^T x_2) = \lambda_1 (x_1^T x_2)$$

$$\Rightarrow x_1^T x_2 = 0 \Rightarrow x_1 \text{ and } x_2 \text{ are orthogonal}$$

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Detecting Eigenpairs $Ax=b$

When given an estimate of an eigenpair $(\hat{\lambda}_i, \hat{x}_i)$, we can check that they are at least close to being a true eigenpair via the idea of a residual:

$$r \triangleq A\hat{x}_i - \hat{\lambda}_i \hat{x}_i \approx 0. \quad A\hat{x}_i = \hat{\lambda}_i \hat{x}_i$$

↳ If we want an eigenvalue estimate that satisfies:

$$|\hat{\lambda}_i - \lambda_i| < \epsilon_{\text{tolerance}}$$

then what can we do?

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The Bauer-Fike Theorem

Theorem 1: Let $A \in \mathbb{R}^{n \times n}$ be a diagonalizable matrix (i.e., $A = P\Lambda P^{-1}$ for some diagonal matrix of eigenvalues $\Lambda \in \mathbb{R}^{n \times n}$ and an invertible matrix $P \in \mathbb{R}^{n \times n}$). Let $(\hat{\lambda}, \hat{x})$ be an approximate eigenpair with residual

$$r = A\hat{x} - \hat{\lambda}\hat{x}.$$

Then there exists an eigenvalue $\lambda \in \sigma(A)$ such that

$$|\hat{\lambda} - \lambda| \leq \frac{\text{cond}_p(P) \|r\|_p}{\|\hat{x}\|_p}.$$

↳ Condition number using p-norm

→ we have $\hat{x}$ and r

→ $\text{cond}_p(P)$ is tricky, more on that later

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$$\det(A - \lambda I)$$

Proof of the Bauer-Fike Theorem

Assume that $\hat{\lambda}$ is not an actual eigenvalue of A (otherwise the proof is trivial). This means that $(A - \hat{\lambda}I)$ is nonsingular, and its inverse therefore exists. That means we can manipulate the expression for the residual as follows:

$$r = A\hat{x} - \hat{\lambda}\hat{x} = (A - \hat{\lambda}I)\hat{x} \Rightarrow \hat{x} = (A - \hat{\lambda}I)^{-1}r$$

$$\text{next, } A = P\Lambda P^{-1}$$

$$\Rightarrow \hat{x} = (P\Lambda P^{-1} - \hat{\lambda}I)^{-1}r$$

$$= [P(\Lambda - \hat{\lambda}I)P^{-1}]^{-1}r = P(\Lambda - \hat{\lambda}I)^{-1}P^{-1}r$$

and then take the norm:

$$\Rightarrow \|\hat{x}\|_p = \|P(\Lambda - \hat{\lambda}I)^{-1}P^{-1}r\|_p \leq \|P\|_p \cdot \|(\Lambda - \hat{\lambda}I)^{-1}\|_p \cdot \|P^{-1}\|_p \cdot \|r\|_p$$

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Proof of the Bauer-Fike Theorem Continued

Note that $(\Lambda - \hat{\lambda}I)^{-1}$ is a diagonal matrix. Therefore, its p -norm is easy to compute:

$$\begin{aligned} \|(\Lambda - \hat{\lambda}I)^{-1}\|_p &= \max_{\|x\|_p \neq 0} \frac{\|(\Lambda - \hat{\lambda}I)^{-1}x\|_p}{\|x\|_p} \rightarrow \text{Induced norm, different from Frobenius norm, it has max } \|x\|_p \text{ to evaluate how this matrix stretch the vector.} \\ &= \max_{\|x\|_p = 1} \left(\sum_{i=1}^n \left(\frac{x_i}{\lambda_i - \hat{\lambda}} \right)^p \right)^{\frac{1}{p}} \rightarrow \text{choose } \lambda_i \text{ so that } |\lambda_i - \hat{\lambda}| \text{ is minimized!} \\ &= \frac{1}{\min_{\lambda_i \in \sigma(A)} |\lambda_i - \hat{\lambda}|} \\ \Rightarrow \min_{\lambda_i \in \sigma(A)} |\lambda_i - \hat{\lambda}| &\leq \text{cond}_p(P) \frac{\|r\|_p}{\|\hat{x}\|_p} \end{aligned}$$

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different from Frobenius norm, it has max $\|x\|_p$ to evaluate how this matrix stretch the vector. compare with $\| \cdot \|_F$, which is "no max" version, $\| \cdot \|_F$ calculate distance.

A More Useful Corollary

and $p=2$
Corollary 1: If A is symmetric, the bound becomes

$$|\hat{\lambda} - \lambda| \leq \frac{\|r\|_2}{\|\hat{x}\|_2} \rightarrow \text{we normalized our estimates } v_k \text{ of the eigenvector for the power method}$$

This is because, by the spectral theorem, a symmetric matrix A can be diagonalized as $A = Q\Lambda Q^T$ where Q is orthogonal ($Q^T Q = I$), and $\text{cond}_p(Q) =$

$$\|Q\|_2 \cdot \|Q^T\|_2 = 1 \cdot 1 = 1$$

now the theorem is actually useful!

Example Eigenpair

Example 1: Consider the symmetric matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

and consider the approximate eigenpair $(1.01, [0.5 \ 0.5]^T)$.

$$\begin{aligned} r &= A\hat{x} - \hat{\lambda}\hat{x} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} - 1.01 \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 \\ 1.0 \end{bmatrix} - \begin{bmatrix} 0.505 \\ 0.505 \end{bmatrix} = \begin{bmatrix} -0.005 \\ 0.495 \end{bmatrix} \\ |\lambda - \hat{\lambda}| &\leq \frac{\|r\|_2}{\|\hat{x}\|_2} \approx \frac{0.495}{0.707} \approx 0.7 \\ \hookrightarrow |1 - 1.01| &= 0.01 \end{aligned}$$

Visualizing the Bound

Example 1 cont.: What happens if we systematically perturb a true eigenpair?

```
using Plots

n_plot = 100
θ = range(0.0, stop=π/2, length=n_plot)
λ = 1.0
dλ = range(0.0, stop = 1.0, length=n_plot)

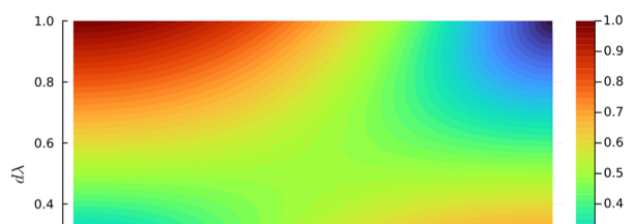
bauer_fike(θ, dλ) = norm((A - (λ+dλ)*I)*[cos(θ); sin(θ)])
bound = bauer_fike(θ', dλ)

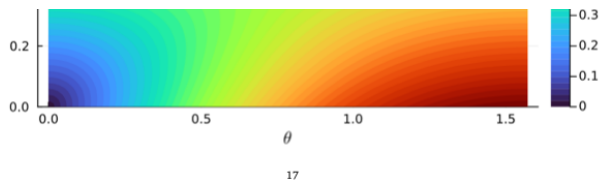
contourf(θ, dλ, bound, fill=true, color=:turbo,
         lw=0.0, aspect_ratio=:equal, levels=50)
xaxis!("\$\\theta\$")
yaxis!("\$d\\lambda\$")
title!("Bauer-Fike Bound")
```

Visualizing the Bound

Example 1 cont.: The resulting contour plot is interesting:

Bauer-Fike Bound





我们想找到最小的eigenvalue, 根据 $Ax = \lambda x, x^T Ax = x^T \lambda x, \lambda = \frac{Ax}{x^T x}$, 然后最小化这个函数; 如果x是一个eigenvector, 那么 λ 就是一个eigenvalue。这个公式可以被解释为**一个方向上的特征值**; 如果x不是一个eigenvector, 那么此时R(x)就是表示为**在x方向上的平均拉伸强度**。然后还有一个乘法性质, 即 $x^T x = \|x\|_2^2$ 乘以 x^T 的原因就是因为我們想让 λ 被求解, 因此希望等式左侧也是一个scaler, 而不是一个vector。

下一页简单的回顾了power method, 其意义是不断地乘以A, 让A最大的eigenvalue所对应的eigenvector不断被放大, 并且让x尽可能地被拉向这个方向; 然后把得到的v带入R(x), 求出估计的 λ , 然后不断迭代。但是在这一张我们更关心的终止条件。

Spectral theorem指出, 如果A是一个real symmetric matrix, 那么:
 $A = Q\Lambda Q^{-1}$, Q是eigenvector组成的矩阵, Λ 是对角矩阵, 对角是A的特征值。这是对称矩阵的好处, Q可以被写成orthogonal matrix, 然后使得 $QQ^T = I$, 所以 $Q^{-1} = Q^T$ 。对于非对称矩阵, 只能写成 $A = PAP^{-1}$, P很不好算。

所以, 我们迭代出来的 λ 和实际差的有多少呢? 我们可以用Bauer-Fike theorem来分析; 如果我们取2 norm, 如果A是real symmetric, 那么 $\text{cond}_2(Q) = 1$, 化简了计算。

关于induced norm和Frobenius norm的区别:

- 算子范数 (Induced Norm, $\|A\|_p$):
 - 定义: 带有 `max` 的定义。
 - 含义: 笔记写道: "evaluate how this matrix stretch the vector" (评估这个矩阵如何拉伸向量)。它关注的是矩阵作为一种变换, 能把向量长度放大的**最大倍数**。
- Frobenius 范数 ($\|A\|_F$):
 - 定义: 笔记写道: "no max version" (没有 `max` 的版本)。
 - 含义: "Calculate distance" (计算距离)。它只是简单地把矩阵里所有元素的平方加起来开根号, 把它当成一个长向量来算长度。